Chapitre 12

RPMD under constrain in 1D system : Energy surface of double well potential (GG) **

We consider a proton in a double well potential

$$V(x) = V_0 \left[\left(\frac{x}{a}\right)^2 - 1 \right]^2 \tag{12.1}$$

The aim of this project is to use the RPMD technique (Ring Polymer Molecular Dynamics) to take into account quantum effects and to compute the barrier energy at a given temperature. We use $a = 0.04$ nm and the value of V_0 will be derived after the choice of the constant $C = \hbar^2/(2ma^2V_0)$ (see complement below). We first consider the case $C = 0.8$ and the temperature $T = 0.2V_0/k_B$.

Reminder : RPMD Hamiltonian

In the Ring Polymer Molecular Dynamics (RPMD) approach, a quantum particle at statistical temperature $\beta = 1/(k_B T)$ is mapped onto a classical ring polymer composed of P replicas (beads) $x_1, \ldots, x_P$ connected by harmonic springs. To sample the canonical distribution at temperature T, each bead is coupled to a Langevin thermostat at the temperature $T_{th} = PT$ via the friction coefficient γ.

The RPMD Hamiltonian is

$$H_P(\mathbf{x}, \mathbf{p}) = \sum_{s=1}^{P} \left[\frac{p_s^2}{2m} + \frac{1}{2} m\omega_P^2 (x_{s+1} - x_s)^2 + \frac{1}{P} V(x_s) \right] \tag{12.2}$$

with

$$\omega_P = \frac{P}{\beta\hbar} = \frac{Pk_BT}{\hbar}, \qquad x_{P+1} \equiv x_1. \tag{12.3}$$

The estimator of the potential energy is

$$\langle V \rangle = \left\langle \frac{1}{P} \sum_{s=1}^{P} V(x_s) \right\rangle. \tag{12.4}$$

1. Write an algorithm to calculate, at a given temperature, kinetic energy, potential energy, and position distributions of beads and of the centroid. Specify the generation of the random force and modifications in the Verlet algorithm to include the thermostat.
2. Write the program.

3. Consider the case $P = 1$ (Trotter number = 1). Choose the optimal values of the time step and the friction coefficient γ.
4. Study the evolution of the potential energy as a function of P. Deduce the optimal value of P.
5. Modify the algorithm to impose a fixed position x_c to the centroid during the dynamics. Include the calculation of the average force f_c on the centroid.
6. Calculate f_c for different values of x_c ranging from 0 to 1. Derive the effective potential at $T = 0.2V_0/k_B$ for the case $C = 0.8$. Comment on the result.
7. Try again (4 to 6) with $T = 0.2V_0/k_B$ and $C = 0.3$. Conclude about the barrier height.
8. Try again (4 to 6) with $T = 0.4V_0/k_B$ and $C = 0.3$. Conclude about the effect of temperature on barrier height for $C = 0.3$.

Complements

Using reduced units, $y = x/a$ and $e = E/V_0$, the stationary Schrödinger equation writes

$$-C\phi'' + (y^2 - 1)^2\phi = e\phi. \tag{12.5}$$

The numerical resolution shows a critical value $C_0 = 0.732$ for which the ground-state energy equals V_0.